

3. Is  $(3, 4, -2)$  a solution of the following system?

$$5x_1 - x_2 + 2x_3 = 7 \quad (1)$$

$$-2x_1 + 6x_2 + 9x_3 = 0 \quad (2)$$

$$-7x_1 + 5x_2 - 3x_3 = -7 \quad (3)$$

$$\text{Eqn 1: } (5 \times 3) - 4 + 2(-2) = 15 - 4 - 4 = 7 \text{ (RHS)}$$

$$\text{Eqn 2: } (-2 \times 3) + 6 \times 4 + 9 \times (-2) = -6 + 24 + (-18) = 0 \text{ (RHS)}$$

$$\text{Eqn 3: } (-7 \times 3) + (5 \times 4) - (3 \times -2) = -21 + 20 + 6 = 5 \neq 7 \text{ (RHS)}$$

$\therefore$  It is not a solution.  $\square$

4. For what values of  $h$  and  $K$  is the following system consistent?

$$2x_1 - x_2 = h$$

$$-6x_1 + 3x_2 = K$$

$$\text{Augmented Matrix: } \begin{bmatrix} 2 & -1 & h \\ -6 & 3 & K \end{bmatrix}$$

New row 2 = Old row 2 + 3x row 1

$$\begin{bmatrix} 2 & -1 & h \\ 0 & 0 & 3h+K \end{bmatrix}$$

The new equations are:

$$2x_1 - x_2 = h$$

$$0x_1 + 0x_2 = 3h+K. \text{ For the system to be consistent, } 3h+K=0.$$

### Exercises 1.1

Solve each system in Exercises 1-4 by using elementary row operations on the equations or on the augmented matrix. Follow the systematic elimination procedure described in this section.

$$1) \begin{cases} x_1 + 5x_2 = 7 \\ -2x_1 - 7x_2 = -5 \end{cases}$$

$$\text{Augmented matrix } \begin{bmatrix} 1 & 5 & 7 \\ -2 & -7 & -5 \end{bmatrix}$$

New row 2 = old row 2 + 2x row 1

$$\begin{bmatrix} 1 & 5 & 7 \\ 0 & 3 & 9 \end{bmatrix}$$

~~Row 2 = old row 2 + 2x row 1~~

Row 3 = Old row 3  $\times \frac{1}{3}$

Row 1 = Old row 1 +  $(-5) \times$  Row 2

$$\begin{bmatrix} 1 & 5 & 7 \\ 0 & 1 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & -8 \\ 0 & 1 & 3 \end{bmatrix}$$

$$\therefore \begin{cases} x_1 = -8 \\ x_2 = 3 \end{cases}$$